

Week 12 Recitation

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ECON2001.01 (Prin. Micro.)

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Market for Factors of production

Labor market

Now we turn our focus to Labor market! (Assumed to be **Perfectly competitive market**, again!)

Labor Demand

- Demander: **Firms** (Employers)
- By employing one more worker,
 - **Marginal Benefit** (= Marginal Revenue):
Value Marginal Product of Labor (VMP_L) = $P \cdot MPL$
 - **Marginal Cost** (of employing one more worker): w (wage)

Labor Supply

- Supplier: Consumers of the final products (Employees)
- By working one more hour, (Hours = Working hour + Leisure time)
 - **Marginal Benefit**: w (wage)
 - **Marginal** (Opportunity) **Cost**: Marginal Benefit from Leisure

Labor Market

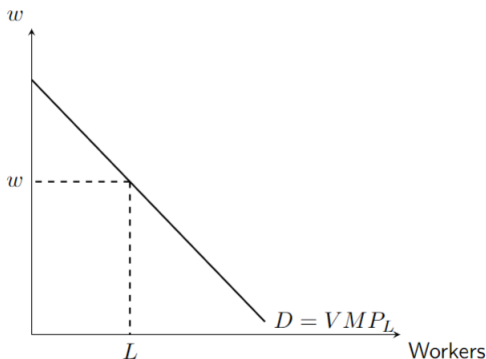
Labor Demand

Demand side optimization

How many workers to employ?

- Marginal cost of labor = wage (w).
 - In the perfectly competitive labor market, the wage is given as w for each firm.
 - If the labor market is not perfectly competitive, the wage can change according to the number of workers.
- Profit maximizing rule:
 $MR = MC \Leftrightarrow VMP_L = w$.
 - If the labor market is not perfectly competitive, the optimal condition still requires $VMP_L = MC$ at the last worker.

Labor Demand Curve



Labor Demand Shifts

- Changes in the output price P
- Technology of the firm
 - If the technology substitutes the labor, labor demand will be reduced.
 - If the technology complements the labor, labor demand will increase. (VMP_L can increase)

Labor Market

Labor Demand

Textbook Example:

Assume that Greg's marginal product is 10 and Jamal's marginal product is 20 and that the bakery earns \$2 per loaf of bread. Further, the market wage for bakers is \$25/hour. Should the bakery hire Greg, Jamal, both, or neither?

Oprah Winfrey's profession of her love for bread shifts out the demand curve, so the bakery now earns \$3 per loaf. Should the bakery hire Greg, Jamal, both, or neither?

Labor Market

Labor Demand

Problem Set #1.

1. (Value of the marginal product of labor; Demand for Labor) Farmer Annie's Ice Cream Shop has calculated the marginal product of labor, and the results are shown in the following table. The wage for workers is \$75 per day, and the pints of ice cream sell for \$5 per pint.

Number of Workers per Day	Marginal Product (Pints of Ice Cream)
0	
1	30
2	35
3	28
4	25
5	20
6	15
7	10

- a. How many workers should Farmer Annie hire?
- b. If the wage for workers increases to \$100 per day, how many workers will Farmer Annie hire?
- c. If the wage for workers is \$75 per day, and the price of ice cream falls to \$3 per pint, how many workers will Farmer Annie hire?

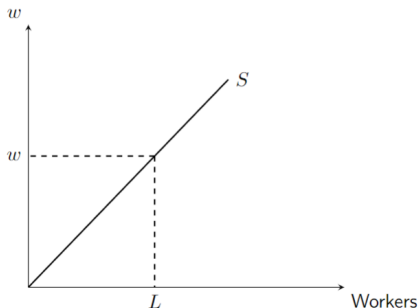
Labor Market

Labor Supply

Supply side optimization

Labor-Leisure Trade-off

- Time we have = Working hours + hours of Leisure
- Opportunity cost of leisure = wage
- Optimization Problem:
 - Consume hours of leisure until the MB from leisure is equal to the opportunity cost of leisure, the wage rate.



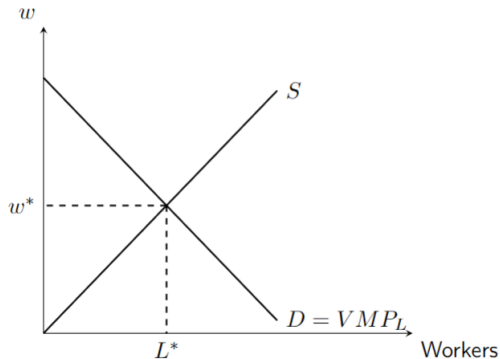
Labor Supply Shifts

- Population changes
 - Market Labor supply changes
- Changes in worker preferences and tastes for leisure
- Opportunity costs
 - E.g., What if the wages in the other industries increase?

Labor market

Labor market equilibrium

The competitive equilibrium in the labor market



Labor market

Labor market equilibrium

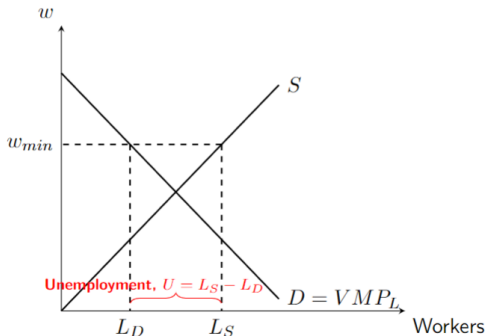
Example: (Problem Set #2.)

Suppose you run a pizza restaurant in a college town, and your primary source of labor is college students working to pay for school. Draw a demand and supply curve for the labor market. Suppose the college receives a large gift and they use it to increase financial aid available to students. Show the effect of this gift on the labor market for local college students.

Labor Market

Government intervention

Minimum wage



DWL from the government intervention?

- CS?
- PS?
- SS?